CAR DEALERSHIP ERP

Departments, Dashboards, Functionalities & Business Workflows

Requirements & Planning Document

A practical planning reference for a multi-category vehicle dealership ERP.

1. Executive Summary

This document outlines the recommended departments, dashboards, functionalities, workflows, and system structure for a comprehensive Car Dealership Enterprise Resource Planning (ERP) platform. It is designed for a dealership that sells new and used vehicles, imports vehicles, accepts trade-ins and swaps, and handles multiple vehicle categories.

The ERP should connect every department through a centralized system while providing role-based dashboards, approvals, reporting, and controlled access. The business can start with fewer departments and split them into dedicated teams as it grows.

2. Recommended Company Departments

No.	Department	Primary responsibility
1	Executive Management	Overall company management and strategic decisions
2	Sales & Marketing	Selling vehicles and generating leads
3	Vehicle Inventory & Stock	Managing vehicle availability, location, and status
4	Procurement & Purchasing	Buying vehicles locally and internationally
5	Importation & Logistics	Shipping, customs clearance, and transportation
6	Trade-In & Vehicle Swap	Managing customer vehicle exchanges
7	Customer Relationship Management (CRM)	Customer records, communication, and follow-up
8	Finance & Accounting	Payments, expenses, accounting, and profitability
9	Human Resources (HR)	Employees, attendance, payroll, and performance
10	Operations & Administration	Daily dealership and branch operations
11	Inspection & Quality Assurance	Vehicle condition checks and sale readiness
12	Service & Maintenance	Repairs, servicing, warranties, and after-sales support
13	Marketing & Media	Advertising, branding, and digital campaigns
14	Legal & Compliance	Contracts, ownership documents, and compliance
15	IT & System Administration	ERP users, security, configuration, and technical operations

3. Executive Management Dashboard

Purpose: Give the CEO, Managing Director, and authorized executives a complete overview of company performance.

Dashboard component	Functionality
Total Vehicles	Total vehicles owned or managed by the company
Available Stock	Vehicles currently available for sale
Total Sales	Sales for today, this month, and this year
Total Revenue	Revenue generated from vehicle sales
Gross / Net Profit	Profitability at the appropriate accounting level
Outstanding Payments	Customers with unpaid balances
Vehicle Acquisition Costs	Purchase and import costs
Active Importations	Vehicles being shipped or cleared
Trade-In Summary	Number and value of vehicles received through swaps
Department Performance	Performance across departments
Employee Performance	Sales, productivity, and employee activity
Expense Overview	Operational and business expenses

- View company-wide financial and operational reports.
- Monitor department and branch performance.
- View sales trends, profit margins, and inventory turnover.
- Approve high-value purchases, discounts, and trade-in valuations within assigned authority.
- Monitor outstanding payments and pending approvals.

• Generate monthly, quarterly, and annual reports.

4. Sales & Marketing Department Dashboard

Purpose: Manage vehicle sales from customer inquiry through negotiation, payment, documentation, and delivery.

Dashboard component	Functionality
Sales Overview	Daily, weekly, monthly, and yearly sales
Available Vehicles	Vehicles available for sale
Sales Leads	Potential buyers and inquiries
Active Deals	Ongoing negotiations
Sales Pipeline	Track inquiry-to-sale stages
Sales Targets	Individual and team targets
Sales Representatives	Staff performance
Pending Reservations	Reserved vehicles
Commission Overview	Sales commission calculations
Follow-Up Reminders	Customers requiring follow-up

• Create sales opportunities and leads.

• Search vehicles by make, model, year, category, price, mileage, and condition.

• Generate quotations and apply authorized discounts.

• Reserve vehicles and create sales orders.

• Record deposits, payment progress, invoices, and receipts.

• Convert leads into customers and schedule appointments.

• Track targets, commissions, lost deals, and reasons.

• Generate sales reports.

Suggested sales workflow: Customer Inquiry → Create Lead → Identify Vehicle → Inspection/Test Drive → Quotation & Negotiation → Reservation → Sales Order → Payment & Documentation → Approval if required → Vehicle Handover → Completed Sale.

5. Vehicle Inventory & Stock Management Dashboard

Purpose: Maintain accurate records of vehicles owned, available, reserved, sold, in transit, or undergoing inspection or repair.

Dashboard component	Functionality
Total Inventory	All vehicles recorded in the ERP
Available Vehicles	Vehicles ready for sale
Reserved Vehicles	Vehicles reserved for customers
Sold Vehicles	Vehicles already sold
In-Transit Vehicles	Vehicles being transported
Under Inspection	Vehicles awaiting inspection
Workshop Vehicles	Vehicles undergoing repairs
Stock Aging	Vehicles held for extended periods
Low-Margin Vehicles	Vehicles with limited expected profitability

• Register acquired vehicles and assign unique stock numbers.

• Store VIN/chassis number, engine number, registration, make, model, year, trim, color, mileage, and fuel type.

• Classify vehicles and upload photos, videos, and documents.

• Record acquisition cost, purchase source, branch, and current location.

• Track vehicle status and movement between branches.

• Prevent double-selling and manage reservations.

- Maintain vehicle history and days in stock.
- Generate inventory valuation and profitability reports.

Recommended vehicle statuses: Draft; Awaiting Approval; Purchased; In Transit; Customs Clearance; Awaiting Inspection; Under Repair; Available; Reserved; Sold; Delivered; Returned; Written Off. A vehicle should have only one primary inventory status at a time.

6. Procurement & Purchasing Dashboard

Purpose: Manage purchases from local suppliers, international dealers, auctions, and manufacturers.

Dashboard component	Functionality
Purchase Overview	Purchase count and value
Purchase Requests	Requests awaiting approval
Purchase Orders	Approved purchase orders
Suppliers	Local and international supplier records
Pending Payments	Supplier payment obligations
Purchase History	Previous vehicle purchases
Supplier Performance	Supplier reliability and quality
Acquisition Costs	Purchase cost analysis

- Register suppliers and vendors.
- Create purchase requests and compare quotations.
- Create purchase orders and submit them for approval.
- Record supplier invoices and payment status.
- Track currencies, exchange rates, and acquisition costs.
- Manage auction purchases and supplier history.
- Generate procurement reports.

Suggested workflow: Purchase Request → Procurement Review → Supplier Quotation → Cost/Profit Analysis → Management Approval → Purchase Order → Supplier Payment → Vehicle Acquisition → Inventory Registration.

7. Importation & Logistics Dashboard

Purpose: Manage international vehicle purchases, shipping, customs clearance, and transportation.

Dashboard component	Functionality
Active Shipments	Vehicles currently being shipped
Pending Importations	Vehicles awaiting shipment
Customs Clearance	Vehicles undergoing clearance
Shipping Costs	Shipping and logistics expenses
Import Documents	Shipping and customs documents
Estimated Arrival	Expected arrival dates
Delayed Shipments	Shipments experiencing delays

- Create import records and link vehicles to purchase orders.
- Record country of origin, supplier, shipping company, clearing agent, vessel/container, and shipment references.
- Track departure, arrival, and shipment milestones.
- Upload bills of lading, invoices, and customs documents.
- Track customs duties, clearing charges, port charges, and demurrage.
- Calculate total landed cost and notify inventory when vehicles arrive.
- Generate importation reports.

Suggested stages: Purchase Approved → Vehicle Purchased → Awaiting Shipment → Shipped → Arrived at Port → Customs Clearance → Cleared → Transported to Dealership → Inspection → Available for Sale.

8. Trade-In & Vehicle Swap Dashboard

Purpose: Manage customers who exchange their existing vehicles for dealership vehicles.

Dashboard component	Functionality
Swap Requests	Customer exchange requests
Pending Valuations	Vehicles awaiting appraisal
Swap Approvals	Deals awaiting authorization
Completed Swaps	Successfully completed exchanges
Trade-In Inventory	Vehicles acquired through swaps
Swap Profitability	Estimated profitability of exchange deals

- Create trade-in requests and record customer vehicle details.
- Upload photographs and documents and schedule inspections.
- Estimate market value and calculate trade-in value.
- Calculate price difference, cash payment, and financing adjustments.
- Submit deals for approval and generate quotations and swap agreements.
- Register acquired trade-in vehicles into inventory.
- Track ownership transfer and generate reports.

Example swap calculation	Amount
Dealership vehicle selling price	$50,000
Customer trade-in valuation	$20,000
Additional amount payable	$30,000

The system should calculate the difference automatically and allow authorized additional charges or discounts.

9. Customer Relationship Management (CRM) Dashboard

Purpose: Maintain customer information and manage relationships throughout the customer lifecycle.

Dashboard component	Functionality
Total Customers	Registered customer count
New Customers	Recently registered customers
Active Leads	Potential customers
Customer Follow-Ups	Pending follow-up activities
Customer Complaints	Unresolved customer issues
Purchase History	Previous purchases
Customer Segments	Individual, corporate, VIP, and other groups

- Register individual and corporate customers.
- Maintain contact details, preferences, and communication history.
- Track inquiries, interests, appointments, and follow-ups.
- Record purchase history, complaints, feedback, and referrals.
- Manage loyalty programs and promotional notifications.
- Support individuals, corporate clients, government organizations, dealers, resellers, VIPs, and fleet customers.

10. Finance & Accounting Dashboard

Purpose: Manage company finances, accounting records, payments, expenses, and profitability.

Dashboard component	Functionality
Total Revenue	Revenue from sales and services
Total Expenses	Operational and business expenses
Gross Profit	Profit before operating expenses
Net Profit	Profit after relevant expenses
Accounts Receivable	Money customers owe
Accounts Payable	Money owed to suppliers
Cash Flow	Cash inflows and outflows
Bank Balances	Balances across company accounts
Outstanding Invoices	Unpaid customer invoices
Financial Reports	Income statement, balance sheet, and related reports

- Manage income, expenses, customer invoices, and supplier payments.
- Manage accounts receivable and payable.
- Track vehicle purchase, shipping, customs, repair, and other costs.
- Calculate vehicle-level margins and profitability.
- Manage bank accounts, cash transactions, and reconciliations.
- Support multiple currencies such as NGN and USD, with exchange-rate records.
- Manage applicable taxes, commissions, and financial approvals.
- Generate profit and loss, balance sheet, cash flow, and other financial reports.

Example vehicle profitability: Purchase price $25,000; shipping $2,000; customs and clearing $3,000; repairs $1,000; other acquisition costs $500; total cost $31,500; selling price $38,000; gross vehicle profit $6,500.

Important: Revenue, gross profit, net profit, and cash flow are different financial measures and should not be treated as interchangeable.

11. Human Resources (HR) Dashboard

Purpose: Manage employees, attendance, payroll, and employee performance.

Dashboard component	Functionality
Total Employees	Company workforce
Present Today	Attendance summary
Pending Leave Requests	Leave awaiting approval
Payroll Overview	Salary and payroll summary
Employee Performance	Staff productivity
Upcoming Birthdays	Employee reminders

- Register employees and maintain profiles.
- Assign departments, job titles, roles, and managers.
- Manage attendance, shifts, schedules, and leave.
- Manage payroll, commissions, bonuses, and salary records.
- Conduct performance reviews and manage employee documents.
- Maintain restricted disciplinary records and generate HR reports.

12. Operations & Administration Dashboard

Purpose: Coordinate daily dealership operations across departments and branches.

Dashboard component	Functionality
Daily Operations	Daily activity summary
Branch Performance	Operational performance by branch
Pending Tasks	Incomplete operational tasks
Vehicle Movements	Transfers and deliveries
Operational Issues	Reported problems
Approvals	Pending operational approvals

- Manage branches, showrooms, and operational tasks.
- Track vehicle movements and coordinate deliveries.
- Manage reservations, schedules, and company assets.
- Monitor branch activity and operational performance.
- Generate operational reports.

13. Inspection & Quality Assurance Dashboard

Purpose: Ensure vehicles meet company standards before display, sale, or delivery.

Dashboard component	Functionality
Pending Inspections	Vehicles awaiting inspection
Completed Inspections	Inspections completed
Failed Inspections	Vehicles requiring repairs
Inspection Reports	Vehicle quality reports
Vehicle Condition	Condition ratings

- Schedule inspections and assign inspectors.
- Use checklists for engine, transmission, brakes, suspension, electrical systems, exterior, interior, tires, safety, and documentation.
- Record mileage, defects, and damages.
- Upload photographs and generate inspection reports.
- Assign repair tasks and approve vehicles for sale after required checks.
- Maintain inspection history.

14. Service & Maintenance Dashboard

Purpose: Manage vehicle servicing, repairs, warranty claims, and after-sales support.

Dashboard component	Functionality
Service Requests	Customer service requests
Active Repairs	Vehicles being repaired
Completed Repairs	Finished service jobs
Pending Parts	Parts awaiting procurement
Warranty Claims	Warranty-related requests
Service Revenue	Service income

- Create appointments, service requests, and repair job cards.
- Assign mechanics and technicians and record diagnoses.
- Manage spare parts, consumables, repair costs, and service history.
- Manage warranty claims and preventive maintenance schedules.
- Notify customers about service progress.
- Generate service invoices and track service revenue.

15. Marketing & Media Dashboard

Purpose: Manage advertising, vehicle promotions, brand visibility, and campaigns.

Dashboard component	Functionality
Active Campaigns	Ongoing advertisements
Campaign Performance	Leads and conversions
Marketing Expenses	Advertising costs
Social Media Activity	Social media performance
Website Leads	Leads from dealership website
Promotional Vehicles	Vehicles currently advertised

- Create campaigns and promotional offers.
- Advertise available vehicles and manage marketing content.
- Track budgets, expenses, leads, and campaign performance.
- Integrate website inquiries and social media leads.
- Calculate marketing return on investment (ROI).

16. Legal & Compliance Dashboard

Purpose: Manage legal documentation, contracts, ownership records, and compliance requirements.

Dashboard component	Functionality
Pending Contracts	Agreements awaiting signatures
Vehicle Documents	Ownership documentation
Expiring Documents	Documents requiring renewal
Legal Cases	Active legal matters
Compliance Tasks	Pending compliance activities

- Manage vehicle ownership documents, sales agreements, swap agreements, and supplier contracts.
- Store importation, insurance, registration, and other vehicle documents.
- Track expiration dates and compliance reminders.
- Maintain legal document history and track disputes.
- Restrict access to sensitive legal records.

For a Nigerian dealership, compliance workflows should be configurable for applicable vehicle registration, ownership transfers, import documentation, taxation, and relevant regulatory authorities. Confirm requirements with qualified local professionals.

17. IT & System Administration Dashboard

Purpose: Manage ERP users, security, configuration, and technical operations.

Dashboard component	Functionality
Total Users	Registered system users
Active Users	Currently active users
Pending Accounts	Accounts awaiting approval
System Activity	Recent user activity
Security Alerts	Suspicious or unauthorized activity
System Health	Application and infrastructure status

- Create and manage user accounts and role permissions.
- Manage department access, branches, settings, and approval workflows.
- Manage integrations, notifications, and system configuration.

- Monitor audit logs, backup status, and security policies.
- Suspend or deactivate accounts and manage master data.

18. Recommended Vehicle Categories

Category	Examples
Executive Class	Mercedes-Benz S-Class, BMW 7 Series
Sport & Performance	Porsche 911, BMW M Series
Premium SUVs	Range Rover, Mercedes-Benz G-Class
Electric Vehicles	Tesla, BYD
Hybrid Vehicles	Toyota Prius, Lexus Hybrid
Luxury Sedans	Lexus ES, Audi A6
Commercial Vehicles	Toyota Hilux, Ford Transit
Trucks & Heavy Vehicles	Volvo Trucks, MAN
Classic & Vintage Cars	Classic Mercedes-Benz, vintage Porsche
Economy Vehicles	Toyota Corolla, Honda Civic
Exotic & Supercars	Lamborghini, Ferrari, McLaren
Buses & Minivans	Toyota Hiace, Mercedes-Benz Sprinter

Each category should support vehicle specifications, pricing, inventory management, sales reporting, and marketing.

19. How Departments Work Together

The ERP should not operate as disconnected dashboards. Departments should share information through a centralized system.

Example vehicle lifecycle: Procurement → Vehicle Purchase → Importation & Logistics → Vehicle Arrival → Inspection → Repair/Reinspection if needed → Inventory → Marketing & Listing → Sales & CRM → Quotation & Reservation → Finance & Payment → Legal Documentation → Delivery → After-Sales Service → CRM follow-up.

Department	Connected departments
Procurement	Finance, Inventory, Importation
Importation	Procurement, Finance, Inventory
Inventory	Sales, Procurement, Inspection
Sales	CRM, Inventory, Finance, Legal
Trade-In	Sales, Inventory, Inspection, Finance
Finance	Sales, Procurement, HR, Operations
HR	Finance, Executive Management
Inspection	Inventory, Service, Operations
Service	Inventory, CRM, Finance
Legal	Sales, Trade-In, Procurement, Finance
Management	All departments

20. Recommended ERP Dashboard Hierarchy

Use three access levels. The system should show data and actions according to the user's role and permissions.

Level	Users	Access
Level 1 — Executive Dashboard	CEO, Managing Director, authorized executives	Company-wide analytics, reports, strategic decisions, and high-level approvals
Level 2 — Department Dashboard	Department managers and authorized staff	Department activities, reports, and departmental approvals

Level	Users	Access
Level 3 — Staff Dashboard	Sales representatives, accountants, inspectors, mechanics, etc.	Assigned tasks, permitted records, and personal performance

Dashboard navigation should be generated dynamically based on the logged-in user's permissions.

21. Recommended ERP Sidebar Navigation

- Dashboard
- Vehicles & Inventory
- Sales & Orders
- Trade-In & Swap
- Customers & CRM
- Procurement
- Importation & Logistics
- Inspection & Quality
- Service & Maintenance
- Finance & Accounting
- Invoices & Payments
- Financial Reports
- HR & Employees
- Branches & Operations
- Marketing & Media
- Legal & Compliance
- Reports & Analytics
- Users & Permissions
- Settings

The sidebar should adapt to role-based permissions so users only see modules they are authorized to access.

22. Essential Shared ERP Features

Feature	Description
Role-Based Access Control (RBAC)	Restrict access according to job roles and permissions
Approval Workflow	Route transactions to authorized approvers
Notifications	Notify users about important activities
Activity Logs	Record user actions and changes
Document Management	Store contracts, invoices, and vehicle documents
Search & Filtering	Search records across modules
Reports & Analytics	Generate departmental and company reports
Multi-Branch Support	Manage multiple dealership locations
Multi-Currency	Support NGN, USD, and other currencies
Audit Trail	Track changes to important financial and operational records
Data Export	Export reports to Excel, PDF, and CSV
Dashboard Customization	Allow users to customize dashboard widgets

23. Important Business Rules

1. A vehicle cannot be sold to two customers simultaneously.
2. A vehicle cannot be sold if it has not been approved for sale.
3. Reservations should have expiration dates and release rules.
4. Discounts above authorized limits require management approval.
5. Vehicle purchases above defined thresholds require approval.
6. Trade-in vehicles must undergo valuation and inspection.
7. Customer payments must be linked to the correct transaction.
8. Vehicle delivery must follow required payment and documentation conditions.
9. Every vehicle should have a complete acquisition and sales history.
10. Financial transactions must maintain an audit trail.
11. Employees should access only information authorized for their roles.
12. Ordinary users should not permanently delete important financial records.

24. Recommended Development Phases

Build the complete architecture with future growth in mind, but implement the most important modules first.

Phase	Modules
Phase 1 — Core ERP	Authentication and permissions; executive dashboard; vehicle inventory; procurement; sales and CRM; trade-in and swap; finance and payments; reports and analytics
Phase 2 — Operations	Importation and logistics; inspection and quality assurance; service and maintenance; branch management; HR and payroll
Phase 3 — Advanced Features	Marketing automation; customer loyalty; advanced analytics; supplier performance analytics; AI-assisted vehicle pricing; AI sales forecasting; business intelligence

The most important foundation is the integration of vehicle inventory, sales, procurement, trade-in, and finance. These modules should be designed together because they represent the core dealership business.

25. Final Recommendation

Start by implementing the core dealership workflow: acquire a vehicle, register it in inventory, inspect it, list it for sale, manage the customer and sale, collect and reconcile payment, complete documentation, and deliver the vehicle.

Build the ERP around a centralized vehicle record, transaction history, role-based permissions, approval workflows, and accounting controls. This foundation will support additional branches, departments, vehicle categories, and advanced functionality as the company grows.